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MILITARY SPECIFICATION

NOISE SUPPRESSOR SYSTEMS, ENGINE TEST STAND A/F32T-2 AND A/F32T-3; FOR TURBOJET AND TURBOFAN ENGINES

1. SCOPE

1.1 Scope. This specification covers demountable noise suppressor systems for use in performance testing of engines mounted on an A/M37T-6 engine test stand. (See 6.5.)

1.2 Classification. The noise suppressor systems shall be of the following types, as specified (see 6.2):

A/F32T-2	For use with J57, J75, and J79 turbojet engines and TF-30, TF-41 turbofan engines
A/F32T-3	For use with J57 and J79 turbojet engines only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-C-596	Connector, Plug, Electrical, Connector, Receptacle, Electrical
QQ-S-781	Strapping, Steel, Flat and Seals

Military

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-3774	Crates, Wood, Open 12,000-and-16,000 Pound
MIL-S-3950	Switch, Toggle, General Specification for

FSC 4920

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MIL-C-5015	Connector, Electric, AN Type, General Specification for
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-8516	Sealing Compound, Polysulfide Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-A-25896	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MIL-T-26289	Test Stand, Aircraft Engine A/M37T-6
MIL-N-83155	Noise Suppressor System, Aircraft Turbine Engine Ground Run-Up, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MIL-STD-1247	Marking, Functions and Hazard Designations of Hose, Pipe, and Tube Lines for Aircraft, Missile, and Space Systems
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS3102	Connector, Receptacle, Electric, Box Mounting
MS24484	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic and Pneumatic (3,000 PSI), Flared Tube
MS33740	Nipple, Pneumatic Starting, 3 Inch ID, Outline Dimensions of

DRAWINGS

Air Force

69E178	Noise Suppressor System A/F32T-2 and A/F32T-3, Design Concept for
69F179	Noise Suppressor System A/F32T-3, Basic Foundation and Utility Plan

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Board of Fire Underwriters

Pamphlet 70 National Electrical Code (Electrical Wiring and Apparatus)

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York, New York 10038.)

National Fire Protection Association

Pamphlet 12A Halogenated Fire Extinguishing Agent System
Pamphlet 12 Carbon Dioxide Systems
Pamphlet 13 Sprinkler Systems
Pamphlet 15 Water Spray Systems

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

American Society of Mechanical Engineers

ASME Boiler and Pressure Vessel Code
Section VIII Unfired Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

National Bureau of Standards

Handbook 46 Code for Protection Against Lightning

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

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3.2 Components. The noise suppressor systems shall consist of the following major components:

Item or System Name	See Requirement
Primary air intake enclosure	3.7
Engine test stand enclosure	3.8
Secondary air intake enclosure	3.9
Augmenter and exhaust muffler	3.10
Water cooling system	3.11
Pumphouse	3.12
Supplementary fuel system	3.13
Electrical system	3.14
Fire protection systems	3.15.

3.3 General specification. The requirements of MIL-N-83155 apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.4 Design. The noise suppressor systems shall be designed to provide a complete and operable facility for performance testing of turbojet and turbofan engines when the engines are mounted on an A/M37T-6 test stand conforming to MIL-T-26289. The general design and layout shall be as shown on Drawing 69E178. The noise suppressors shall be designed to utilize the foundations and utilities shown on Drawing 69F179. The noise suppressor systems shall be designed as demountable systems in accordance with MIL-N-83155 and as follows:

a. The A/F32T-2 noise suppressor system shall be designed for use with the following engines:

(1) J57-P turbojet engines, model dash numbers -7, -13, -19, -21, -23, -29, -37, -39, -43, -55, and -59

(2) J75-P turbojet engines, model dash numbers -5, -17, and -19

(3) J79-GE turbojet engines, model dash numbers -3, -7, -11, -15, -17, and -19

(4) TF30-P turbofan engines, model dash numbers -1, -3, -7, -9, and -100

(5) TF41-A turbofan engines, model dash number -1

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b. The A/F32T-3 noise suppressor system shall be designed for use with the following engines:

(1) J57-P turbojet engines, model dash numbers -7, -13, -19, -21, -23, -29, -37, -39, -43, -55, and -59

(2) J79-GE turbojet engines, model dash numbers -3, -7, -11, -15, -17, and -19

The noise suppressor systems shall be designed for left- and right-hand operation (left- or right-hand side of the noise suppressor). All system components and connections within and between the pumphouse and the control cab, fuel trailer, and engine runup trailer of the A/M37T-6 test stand shall be designed to permit either location without additional modifications to any part of the system. Except for the Government-furnished items (see 3.19), all components necessary to provide a complete functional system shall be provided (see 6.2).

3.4.1 Aerodynamic and thermodynamic limitations. The gas stream flow velocities and temperatures listed in table I shall not be exceeded in the design of the noise suppressor system components. The maximum allowable flow velocities stated are for smooth, diffuse gas flows at grazing incidence to the facing sheet of the panel. A maximum cell depression of 2 inches of water will be permitted in the test stand enclosure for all engines specified herein when operated at full military and afterburner power. The depression shall be measured at the location specified in 3.8.15. The noise suppressor system shall be so designed that the engine being tested will not be exposed to a backpressure.

3.5 Construction. Table I illustrates typical component construction that may be used in the noise suppressor systems. The component minimum metal thickness requirements are based on the use of mild steel and have been established as the minimums acceptable to satisfy construction requirements only and in many cases may have to be increased to permit the required acoustical performance to be achieved of particular components. All acoustical panels shall be provided with separators, reducing the height of unsupported acoustical material to 24 to 36 inches, to prevent settling of the acoustical fill materials. Where material or overall thicknesses and acoustical fill densities are not indicated in table I, these values shall be determined to satisfy the overall requirements for the noise suppressor systems.

3.6 Performance

3.6.1 Acoustical performance

3.6.1.1 Far-field acoustical performance. The noise suppressor in operation with the engines associated with each system identified in 3.4 shall provide

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TABLE I. Component Aerodynamic, Thermodynamic, and Construction Criteria

COMPONENT	ACCEPTABLE TYPES OF CONSTRUCTION	MAX. FLOW VELOCITY (Ft/Sec)	MAX. MIXED GAS TEMP. (°F)
Exhaust plenum and augmenter walls*	5/16-inch solid inner shell, firebox quality steel, or equal Reinforced with structural steel members 11-gage External cover 4-inch thick 6 pcf acoustical fill		
Exhaust silencer shell	5/16-inch solid firebox quality steel, or equal. If acoustical treatment is provided on the inner surface of the shell, the construction shall be equivalent to that shown below for the exhaust silencer dissipative noise control elements. If lagging is provided on the outer surface of the shell, the external cover shall be 11 gage.		
Exhaust silencer dissipative noise control elements	3/16-inch perforated facing S.S. wire screen Glass fiber cloth Acoustical fill	180	600
	3/16-inch perforated facing porous metal separator 14-Gage perforated facing S.S. wire screen Glass fiber cloth Acoustical fill	250	700
Exhaust muffler resonators	1/4-inch firebox quality steel, or equal	300	800
Test stand enclosure walls and ceiling	16-gage perforated facing Glass fiber cloth 1/4-inch external cover 8-inch thick 4-pcf acoustical fill	90 Ahead of engine intake 5 Past engine	
Primary and secondary air intake enclosure walls and ceilings	16-gage perforated facing Glass fiber cloth 3/16-inch external cover 8-inch thick 4 pcf acoustical fill	90	
Primary and secondary air intake silencer shells	16-gage perforated facing Glass fiber cloth 3/16-inch external cover 4-inch thick acoustical fill	90	
Primary and secondary air intake silencer dissipative noise control elements	16-gage perforated facing Glass fiber cloth Acoustical fill	90	

* NOTE: If augmenter exterior walls are not exposed to the atmosphere, 5/16 inch solid firebox quality steel, or equal, will be considered acceptable construction

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grade II acoustical performance as specified in MIL-N-83155, when the sound pressure levels measured at 250 feet from the unsuppressed test stand mounted engines have values up to and including those specified in table II.

3.6.1.1.1 Consideration of possible error in measuring far-field acoustical data. To account for possible instrumentation error, ground reflection, temperature gradients, and other possible errors in recording sound pressure levels during the far-field acoustical performance evaluation, the noise suppressor system shall be considered to have satisfied grade II performance if the measured levels do not exceed grade II levels by more than 4 db. If an unsuppressed far-field acoustical evaluation is performed under the conditions identified in 3.6.1.1.2, the measured sound pressure levels shall be considered as not exceeding table II values if they are no more than 4 db higher.

3.6.1.1.2 Unsuppressed acoustical survey. If grade II acoustical performance, after consideration is given to possible error in measurement as specified in 3.6.1.1.1, is not provided by the noise suppressor system at one or more measurement points, an unsuppressed acoustical survey may be performed. The unsuppressed survey shall be performed as specified by the procuring activity (see 6.2) if the supplier considers the failure to be caused by the engine used in the test producing a higher sound pressure level at the measurement point(s) in question than the table II value(s) and requests that the unsuppressed survey be performed. The surface over which the unsuppressed survey is conducted shall be as hard or harder than the surface over which the suppressed survey was performed. Corrections for molecular absorption as specified in MIL-N-83155 shall be applied to the unsuppressed data. The noise suppressor system shall be considered acoustically acceptable if grade II levels are not exceeded by an amount greater than the measured unsuppressed levels exceed the values in table II after both measured suppressed and unsuppressed data are given the consideration for possible error in measurement as specified in 3.6.1.1.1.

3.6.1.2 Near-field acoustical performance. The near-field acoustical requirements of MIL-N-83155 are not applicable to these systems.

3.6.2 Effect on engine performance. The engine performance parameters to be used in determining the effect of the noise suppressor system on the engine shall be those parameters recorded during engine performance testing as listed in the appropriate engine technical orders (see 6.4).

3.6.3 Operational capability. The noise suppressor systems shall permit operation of the engines specified in 3.4 up to and including maximum afterburning power levels, where applicable.

3.7 Primary air intake enclosure. The primary air intake enclosure shall provide an acoustically treated path for the airflow to the test engine

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TABLE II. Maximum Unsuppressed Sound Pressure Levels Associated With the Turbojet and Turbofan Engines to be Used During the Acoustical Evaluation

Position Rel. To Engine Exhaust Nozzle		Sound Pressure Level (In DB Ref. 0.0002 Dynes/CM^2)							
		Octave Band Center Freq. Hz.							
Angle In Degrees (0° Forward)	Distance In Feet	63	125	250	500	1,000	2,000	4,000	8,000
0	250	106	109	108	106	104	103	102	99
10	250	107	112	109	108	106	104	103	100
20	250	108	113	112	111	108	105	104	101
30	250	108	113	112	112	108	106	107	102
40	250	109	113	112	113	109	108	108	104
50	250	109	113	113	113	110	108	109	105
60	250	109	113	116	115	112	110	112	109
70	250	110	115	117	117	114	111	114	111
80	250	111	117	117	118	115	114	115	111
90	250	113	118	119	120	118	116	117	114
100	250	116	119	122	123	120	116	117	115
110	250	121	123	125	125	123	119	118	118
120	250	127	132	132	127	125	121	120	120
130	250	130	134	133	129	127	123	123	123
140	250	130	134	133	129	127	123	123	123
150	250	128	129	128	124	121	117	116	114
160	250	126	127	123	119	115	111	109	105
170	250	120	121	117	112	108	104	102	98

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and sides will not sag or be deformed. The tank shall be mounted on reinforced

compressor inlet. The air shall enter the intake stack vertically. If required, the primary air intake shall have drainage provisions to prevent the accumulation of water and ice. The inlet stack and any required acoustical treatment shall be designed to accommodate the maximum primary airflow requirements for all of the engines listed in 3.4. Hinged double doors opening to give an unobstructed access area of 10 by 10 feet shall be provided in the front wall to permit movement of the engine test stand and engines on transporter trailers in and out of the test stand enclosure area.

3.8 Engine test stand enclosure. The area between the primary and secondary air intake enclosures shall contain the A/M37T-6 engine runup trailer and shall be completely enclosed by an acoustically treated and weatherproof structure. The internal dimensions of the enclosure shall be 14 feet wide by 12 feet high. The roof shall be sloped to prevent the accumulation of water. The enclosure internal walls in the vicinity of the engine bellmouth shall be uniform and free of protuberances or indentations. To provide essentially laminar flow at the engines' inlets, the distance between the primary air intake stack and the engine inlet bellmouth of any engine listed in 3.4 shall be not less than 12 feet. The acoustical treatment shall withstand temperatures up to and including 140°F, and the acoustical materials shall not support combustion. Personnel protection from engine explosion shall be provided by reinforcing the enclosure wall in the areas shown on Drawing 69E178 up to a height of 8 feet using a 5-ply nylon bulletproof mat, sandwiched between the outer 1/4-inch steel surface and a second internal 1/4-inch plate. The inside wall face sheets shall be free of perforations up to a height of 6 inches above the floor level, to allow washdown of the enclosure with water without contamination of the acoustical materials in the walls. The design of the walls shall prevent the accumulation of water or condensate.

3.8.1 Observation windows. A sealed, double-pane, shatterproof, glass observation window shall be provided in each enclosure wall at the positions shown on Drawing 69E178. The windows shall be in line with the observation window of the A/M37T-6 control cab, without restricting the view of the engine under test. A detachable seal shall be installed on one side of the enclosure to provide a weatherproof connection with the control cab. A detachable rain shield shall be installed above the enclosure observation window not utilized with the control cab for the protection of personnel using this observation position. Desiccant shall be used between the glass panes of the observation windows to prevent vapor condensation. The test stand enclosure observation window shall be a minimum size of 18 by 60 inches and so located to mate with the A/M37T-6 engine control cab observation window having dimensions of 12 inches high by 48 inches long with the horizontal centerline 71 ±1 inch above ground level.

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3.8.1.1 Observation window step. A raised step shall be provided for use at the observation window not being used with the control cab. The height of the step shall be such that the distance from the top of the step to the centerline of the observation window will be 5 feet 6 inches (see Drawing 69E178).

3.8.2 Side access doors. Four personnel access doors (two on each side) shall be provided in the enclosure walls as shown on Drawing 69E178. Each door shall be a minimum of 80 inches high by 36 inches wide and shall be equipped with an 18- by 24-inch double-pane, shatterproof observation window, located with the 24-inch dimension horizontal and centered 66 inches above ground level. The doors shall be so constructed that they open outwards, are readily accessible, unobstructed, and quick opening. The door latches shall be of the emergency exit type and shall operate from the inside or outside by a single motion of the hand. The door sills shall be raised above the floor of the enclosure to prevent spilled fuel or washdown water from flowing out of the enclosure and to facilitate installation of the washdown system.

3.8.3 Test stand enclosure rollup door. To provide a windbreak for ground personnel working in the engine enclosure, a rollup door shall be installed at the beginning of the enclosure as shown on Drawing 69E178 to isolate the primary air intake area. The rollup door shall be constructed of interlocking, lightweight metal slats, coiled on a pipe of sufficient size to carry the door load with a deflection of not more than 0.03 inch per foot of door width. Structural steel guides shall be provided to retain the door in the correct position when lowered.

3.8.3.1 Rollup door operation. The door shall be so designed that it can be manually or electrically raised and lowered. A continuous chain shall be provided for manual operation. The chain shall be stored in the motor hood. Electrical operation shall be accomplished by a high-starting, torque-type motor provided with thermal protection.

3.8.3.2 Electrical control circuit. A NEMA-7 pushbutton switch shall be provided on the wall on each side of the rollup door. Automatic limit switches shall be provided to break the electrical circuit at the door travel limits. The switch shall contain close, open, and stop pushbuttons. The close button shall be a continuous pressure button so that if pressure is removed, the door will stop downward travel and revert to the fully open position. The open button shall be an instantaneous pressure button that will open the closed and partially closed positions of the door to the fully open position. The stop button shall be an instantaneous pressure button that will stop the door in any position.

3.8.3.3 Safety device. A safety device shall be provided across the bottom of the door that will automatically stop the downward travel of the door upon contact with an obstruction in the opening and reverse the door travel to the fully open position.

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3.8.4 Hoist. A hand-operated chain hoist shall be provided and shall have a minimum lifting capacity of 2,000-pounds. The hand chain pull required to lift a 2,000-pound load shall not exceed 80 pounds. The hoist shall be equipped with a noncorrosive-nonsparking hand and load chain. The hoist hook shall be equipped with a safety latch.

3.8.4.1 Hoist trolley and rail. The hoist specified in 3.8.4 shall be mounted on a trolley that can be manually traversed on an overhead rail along the test stand centerline as shown on Drawing 69E178. The hoist rail shall be curved from the centerline to provide a storage position for the hoist on the inside wall of the test stand enclosure. Provision shall be made to restrain the hoist against the inside wall of the test stand enclosure. The rail may be extended to the rear of the test stand into the secondary air enclosure for the distance required for installation of the peculiar engine adapter ducts. The maximum horizontal force required to move the hoist trolley on the rail with the hoist supporting 2,000 pounds shall not exceed 40 pounds.

3.8.4.1.1 Safety factor. The hoist trolley and rail assembly shall be designed for a safety factor of 3 based on the yield strength of the material.

3.8.5 Engine runup trailer installation. The A/M37T-6 engine runup trailer shall be installed with a modified thrust and side restraint system (see 6.2). The trailer rail faces shall be adjusted to 40 ±1/16 inch above floor level to permit the transfer of engines utilizing standard engine transporter trailers.

3.8.5.1 Hydraulic and electrical junction boxes relocation. The hydraulic and electrical junction boxes, normally located on the engine runup trailer, shall be relocated above the engine as shown on Drawing 69E178. The boxes shall be supported by simple frame superstructures fixed to the enclosure walls and ceiling, and shall be attached to the superstructure in the same manner as they are attached to the engine runup trailer.

3.8.5.2 Raceways. Raceways shall be provided to house all interconnecting electrical wiring, fluid lines, and flexible hoses between the quick-disconnect terminal boxes and the system components located outside the enclosure. The raceways shall be accessible for the initial placement and subsequent replacement of all wiring and flexible hoses required. The flexible hoses and electrical wiring shall be in different raceways. The raceways shall be free of sharp edges that would damage the items contained therein.

3.8.5.3 Runup trailer fuel system connection. In lieu of the fuel hose supplied with the A/M37T-6 test stand system, a permanent fuel pipeline shall be provided as part of the foundation, with its terminations 1 foot above

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grade at the locations shown on Drawing 69E178. Two hoses with adapters shall be provided to interconnect between the fuel pipe terminations and the fuel connections on the engine runup and auxiliary equipment trailers. The fuel hose shall be Aeroquip part No. 390A40D, or equal.

3.8.6 Pneumatic starting duct system. A pneumatic starting duct system consisting of a 3-inch ID stainless-steel pneumatic duct and necessary end fittings shall be provided. The duct inlet shall be located outside the enclosure wall at the position shown on Drawing 69E178, terminating at a height 4 feet above ground level. An MS33740 nipple shall be provided in the inlet to allow connection with the Type MA-1A or A/M32A-60 air start cart supply hose. The duct outlet shall be adjacent to the electrical junction box as indicated on Drawing 69E178 and shall be provided with a quick-disconnect fitting to mate with the MS33740 nipple on the connecting pneumatic hose to the engine.

3.8.7 Electric load bank provisions. Two terminal boxes shall be provided to connect the A-1A electric load bank to the engine under test. One box shall be mounted on a simple frame superstructure fixed to the enclosure walls and ceiling, immediately to the rear of the relocated electrical junction box (see 3.5.8.1). The other terminal box shall be weatherproof and mounted on the outer wall of the enclosure at the location shown on Drawing 69E178 with a centerline height of 4 feet above ground level. Each terminal box shall have nine 1/2-inch terminal studs, type 13 UNC-2, with a stud length of 1-1/2 inches, constructed of corrosion-resistant material, and provided with terminal nuts and washers. Corresponding terminal studs in the two terminal boxes shall be connected by electrical conductors, minimum size AWG MCM000, capable of carrying 40 kva. To eliminate the hazard of electric shock, a means shall be provided to protect personnel from contact with the terminal studs. To provide an interconnecting control cable for generators mounted on the test engine, an additional electrical connector shall be mounted on each terminal box. A connector conforming to MS3102-20-16S shall be provided and mounted on the terminal box in the test stand enclosure, and a connector conforming to MS3102-20-16P shall be provided and mounted on the terminal box located outside the test stand enclosure. Corresponding terminals of these connectors shall be interconnected using size 14 wire.

3.8.8 Intercommunication extensions. The intercommunication system of the A/M37T-6 test stand shall be extended to allow communication between the control cab operator and other test personnel inside and outside of the test stand enclosure. Six intercommunication positions, two inside and four outside of the enclosure, shall be provided at the positions shown on Drawing 69E178 allowing the use of headsets and microphones. A connector conforming to PTOOE-10-6S shall be provided at each intercommunication position and shall be protected from the weather where located outside of the enclosure. A dust

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and moisture seal protective cap shall be provided for each connector and shall be permanently attached to the connector. The interconnecting wiring circuits shall be brought to the control cab to allow connection to the two socket terminals on the external terminal panel (see figure 1).

3.8.9 Electrical power outlets. Eight duplex female receptacles shall be provided at the positions shown on Drawing 69E178 for 120-V, 60-Hz, single-phase, 20-amp service. The receptacles shall conform to W-C-596, style X, and shall incorporate a weatherproof spring cover.

3.8.9.1 Switch. A circuit-breaker combination switch shall be provided to supply power to the electrical power outlets. The switch shall be mounted adjacent to the light switches specified in 3.8.14.1.

3.8.10 Lightning protection system. A lightning protection system conforming to Handbook 46 shall be provided. Air terminals shall be constructed of solid copper, 1/2 inch diameter below the taper and 24 inches in height, and shall be placed on the corners of the primary and secondary air intake stacks as shown on Drawing 69E178, and on the exhaust muffler as necessary. The interconnecting cable to connect with the grounding system shall have a minimum diameter of 3/8 inch and shall weigh not less than 187.5 pounds per thousand feet. The size of wires in stranded cable shall be not less than 14 AWG (0.064 inch).

3.8.11 Warning lights. To warn personnel that engine testing is in progress, a red warning light shall be installed on the exterior wall above each of the five access positions to the test enclosure. The lights shall be operated through a control circuit on the control panel in the control cab. A warning light shall also be included in the control panel to indicate to the test operator when any door is opened after the circuit to the exterior warning lights has been switched on.

3.8.12 Interior lighting. A sufficient number of explosion-proof, color improved mercury vapor floodlights shall be provided to illuminate the engine surface to a minimum of 50 footcandles while mounted on the A/M37T-6 engine runup trailer. The engine surface shall be considered a cylinder 4 feet in diameter, less the 30° segment on each side of the lower vertical centerline, extending 3 feet forward and 5 feet aft of the engine runup trailer with the engine horizontal centerline 55 inches above floor level. The lights shall be mounted on the enclosure walls on each side of the engine runup trailer with the lowest point of each floodlight a minimum distance of 9 feet above floor level. Each light shall be adjustable in the horizontal and vertical directions through 180°.

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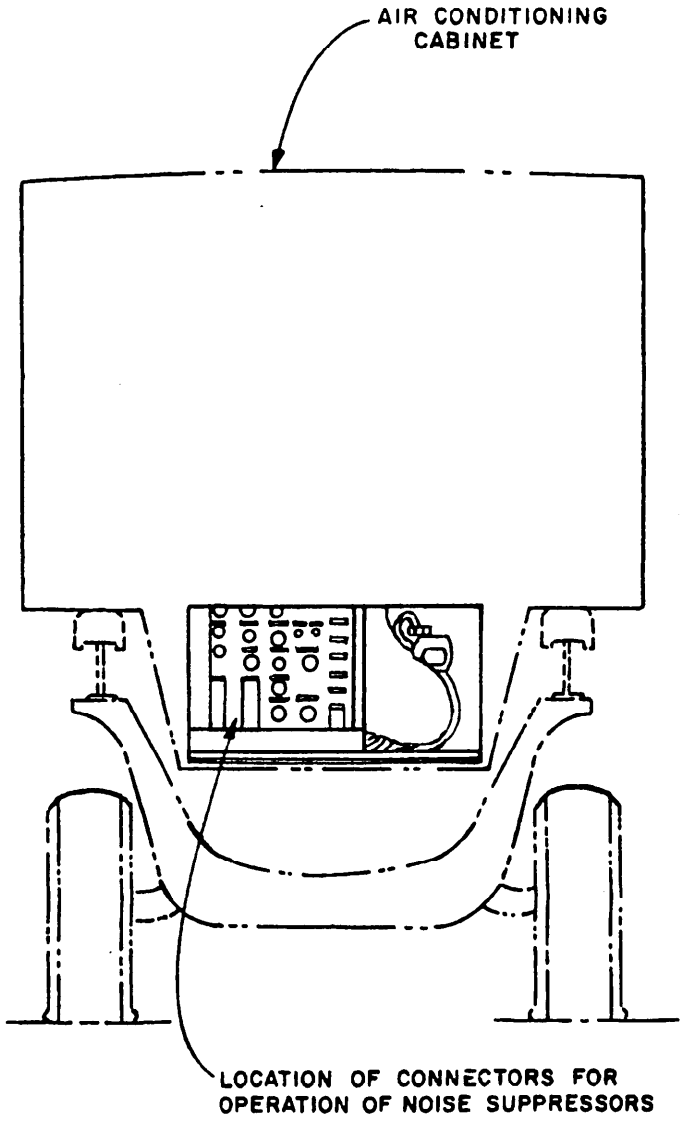


FIGURE 1. A/M37T-6 Control Cab - Exterior Terminal Box

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3.8.12.1 General illumination. The primary and secondary air intakes and test stand enclosure shall be generally illuminated by the provision of wall-mounted lights to provide a minimum illuminance of 25 footcandles at all positions 30 inches above floor level. The lights shall be a minimum distance of 9 feet above floor level and shall be explosion-proof.

3.8.13 Exterior lighting. A sufficient number of color improved mercury vapor floodlights shall be provided to illuminate the ground surface, for a distance of 20 feet on each side of the primary air intake, engine test stand enclosure, and secondary air intake enclosure, to a minimum of 2 footcandles. The ground surface, for a distance of 20 feet in front of the primary air intake, shall be illuminated to a minimum of 3 footcandles. The floodlights shall be mounted on the exterior of the noise suppressor components at a minimum distance of 9 feet above ground level.

3.8.14 Light switches

3.8.14.1 Floodlights. A maximum of two interior floodlights shall be mounted on the same wall and shall be operable from one switch. Circuit-breaker combination type switches shall be provided for all light circuits. The switches for the exterior lights shall be located on the exterior wall adjacent to the engine double access doors in the primary air intake. The switches for the interior lights shall be located on the exterior wall near the rear side access door closest to the control cab.

3.8.14.2 General illumination. General illumination light switches shall be located on the exterior wall near the rear side access door closest to the control cab.

3.8.15 Enclosure air pressure measurement. A reference pressure measuring system shall be installed to measure the atmospheric depression in the test stand enclosure during operation of the engine at all power levels. A static-type pressure probe shall be mounted on the enclosure wall at a height of 5 feet above floor level and in the plane of the engine air inlet. The probe shall be interconnected with the barometric pressure instrument already mounted in the control cab through the external adapter fitted to the instrument pressure line.

3.8.16 Enclosure floor cleaning. To facilitate cleaning the test stand enclosure floor, a water spray system shall be mounted at floor level on the enclosure walls. The water supply to the spray system shall be regulated by a hand control valve so located inside the enclosure as to permit operation by personnel standing outside the rear personnel access door on the same side as the pumphouse specified in 3.12. A 3/4-inch bibcock (for use with a standard water hose) shall be installed upstream of the water spray system control valve.

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3.8.16.1 Drainage. Drainage of the noise suppressor system shall match to the drainage outlets provided in the foundation as shown on Drawing 69F179.

3.8.17 Noise suppressor control panel. A control panel for operation of the noise suppressor system shall be provided and installed in the A/M37T-6 control cab. This panel shall contain all controls, warning indicators, and switches for operation of the noise suppressor system. The dimensions for panel A, open area B, and the location of the air conditioning inlet duct are specified on figure 2. The control panel shall be designed to replace the blank panel (A on figure 3) and utilize the open area (B on figure 3) to the left of the operator's console of the A/M37T-6 control cab or utilize the open area (B as shown on figure 3) to the left and right of the operator's console of the A/M37T-6 control cab. Electrical connections inside the control cab shall be made to a terminal strip mounted behind the left side of the operator's console. The terminal strip is wired to the connector in the external terminal box of the control cab.

3.8.17.1 Control panel external connection. External electrical connection to the control panel shall be by means of an MS3102-36-7S connector located in the external terminal box of the control cab (see figure 1). This connector shall also be used for interconnecting the fire suppression control unit (see 3.15).

3.8.17.2 Control panel controls and indications. The noise suppressor control panel(s) shall contain the following warning indications and controls:

a. Low water level red warning light with buzzer

b. Low water flow rate red warning light with buzzer

c. High exhaust muffler temperature red warning light with buzzer

d. Roll-up door full up position green light

e. Test stand enclosure personnel and primary air inlet double doors open red warning light

f. 60-Hz electrical power failure red light

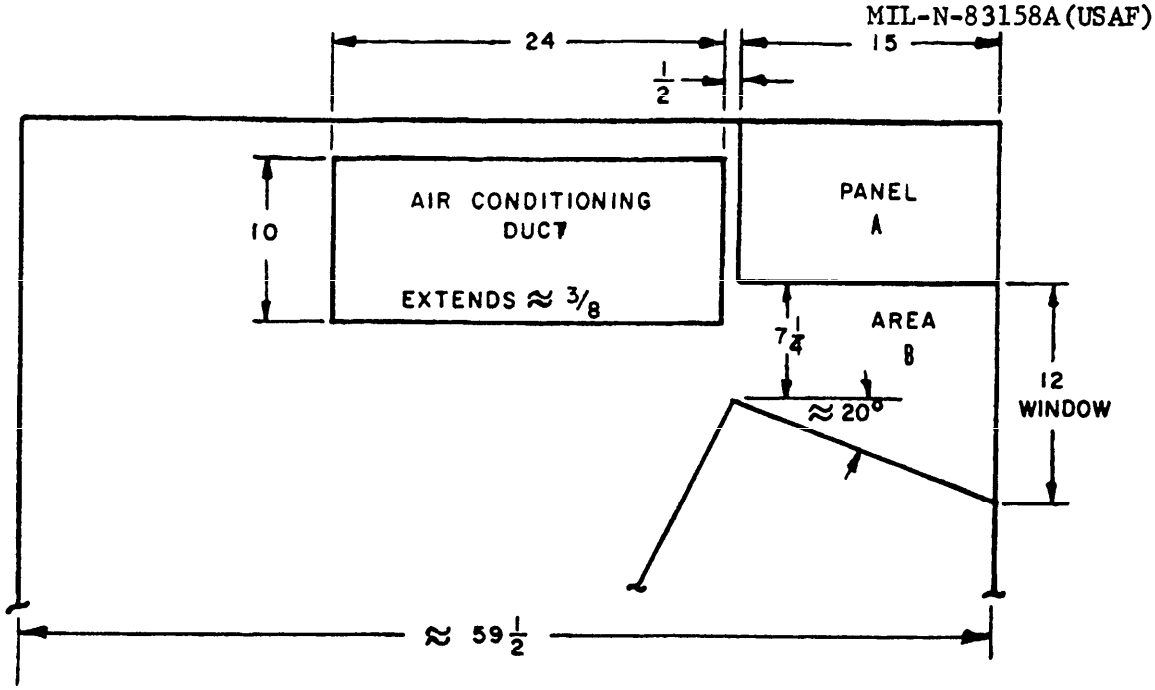
g. Water pump on/off switch with green indicator light for pump operation

h. Switch to activate the warning lights above the personnel doors in the test stand enclosure and above the double doors in the primary air inlet

i. Water control switch(es) with green indicator light(s) for water flow

j. Switch(es) to test the operation of the overtemperature, water flow rate, low water level and electrical power loss warning system.

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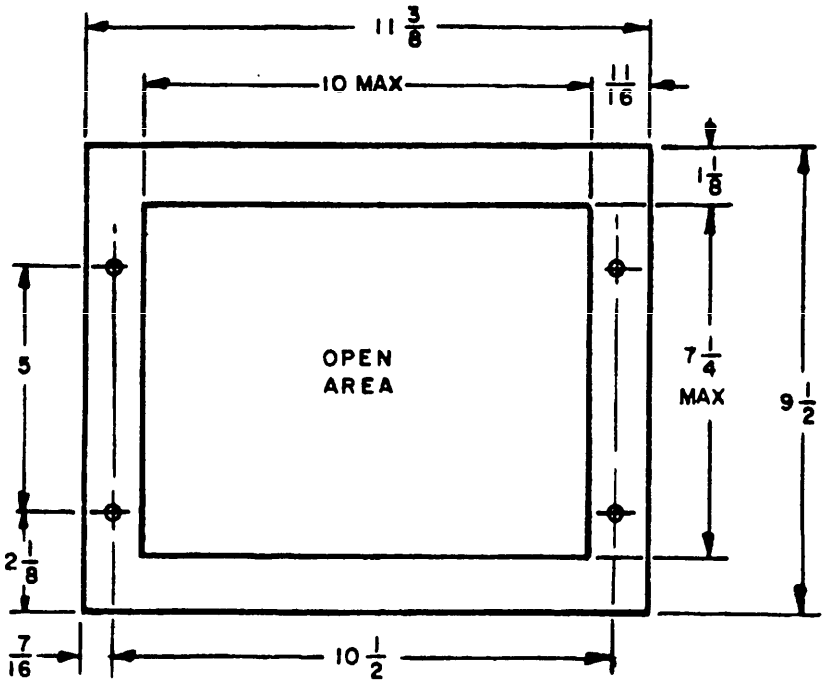
CONTROL CAB
END VIEW LOOKING INSIDE FROM DOORWAY

FACE PANEL
0.125 X 9 1/4 X 11 3/8

RETAINING RING
P/N 82-32-101-17

FASTENER
P/N 82-11-260-17

MANUFACTURER CODE
94222

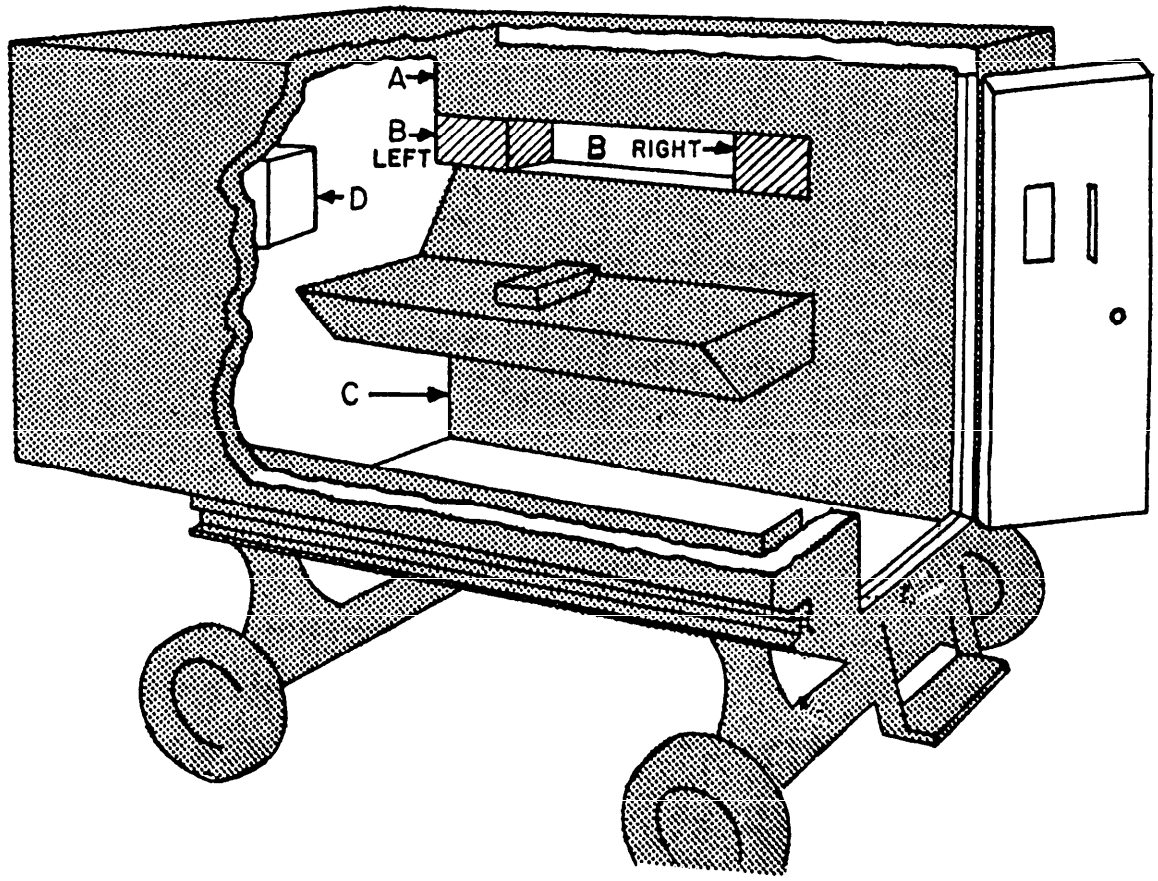


PANEL A
FRONT VIEW

NOTE:
DIMENSIONS IN INCHES

FIGURE 2. Dimensions for Control Panel Areas

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A – BLANK PANEL
B – OPEN AREA
C – OPERATOR'S CONSOLE
D – FIRE SUPPRESSION CONTROL UNIT

FIGURE 3. A/M37T-6 Control Cab

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3.8.18 Septum wall. A septum wall shall be provided to separate the test stand enclosure from the secondary air intake enclosure. The septum wall shall be mounted aft of the A/M37T-6 engine runup trailer. The engine runup trailer thrust restraining I-beams may be used without modification to restrain the septum wall.

3.8.18.1 Personnel access doors. Two personnel access doors shall be provided in the septum wall for personnel entrance into the secondary air enclosure. One door shall be positioned on each side of the A/M37T-6 engine runup trailer near the outside walls of the enclosure. Each door shall be so constructed that it will open into the test stand enclosure; be hinged on the enclosure wall side; be readily accessible; and be equipped with emergency exit, single-motion door latches.

3.8.18.2 Engine adapter door(s). Engine adapter door(s) shall be provided to seal around the exhaust nozzle sections of the engine(s) being tested on the A/M37T-6 engine runup trailer. The engine adapter door(s) shall be permanently mounted to or capable of installation on or removal from the septum wall within 10 minutes. The engine adapter door(s) assembly shall be locked into position for engine testing by personnel standing on ground level. The engine adapter door(s) required for each noise suppressor system shall be as specified by the procuring activity (see 6.2). If the engine adapter doors are not permanently mounted to the septum wall, the doors shall be installed by the hoist or shall meet the maximum weights specified in 3.8.18.2.1.

3.8.18.2.1 Weight limitations. If the engine adapter door(s) are manually installed, each engine adapter door assembly or portion thereof shall have a maximum weight of 54 pounds if it must be lifted to shoulder height and 76 pounds if it must be lifted to waist level and shall be capable of installation and removal by two personnel in the 5th percentile standing on ground level. The 5th percentile is specified in MIL-STD-1472.

3.8.18.3 Engine adapter tube(s). When specified (see 6.2), an engine adapter tube shall be provided for the J57 non A/B engines and for the TF41 engine. The engine adapter tube(s) shall be so designed that it can be installed or removed from the septum wall within 10 minutes. The adapter tubes shall be installed by the hoist or shall meet the maximum weights specified in 3.8.18.2.1.

3.8.19 Emergency breakaway switch cable. A cable shall be provided to interconnect the electrical terminal box and the emergency breakaway switch cable mounted on the engine runup trailer. This cable shall be in series with the commercial power source and fire protection system so that any commercial power failure or fire will immediately cut off power to the engine runup trailer and stop water flow to the noise suppressor water spray system.

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3.9 Secondary air intake enclosure. The secondary air intake enclosure shall provide an acoustically treated path for the turbojet or turbofan engine exhaust gases and the augmented air, between the test stand mounted engine and the augmenter. The enclosure shall incorporate secondary air inlet acoustical treatment where necessary. The augmented air shall satisfy the tertiary air requirements for the TF-30 turbofan engines in addition to the normal augmented cooling air design requirements. The concrete foundations of the enclosure shall not come into direct contact with the exhaust gas stream and, if necessary, shall be protected by the installation of a steel floor covering.

3.9.1 Secondary air inlet observation windows. An observation window shall be provided on each side of the secondary air inlet. Each window shall be 18 inches high by 24 inches wide with a window horizontal centerline 66 inches above ground level. The windows shall be sealed, double-pane, shatterproof glass with desiccant provided between the glass panes. The windows shall be located to permit viewing of the aft section of the engine being tested.

3.10 Augmenter and exhaust muffler. The augmenter and exhaust muffler shall be correctly sized and designed to meet the aerothermodynamic and acoustical requirements for the two independent systems specified in 3.4 and shall be cooled by water and augmented air. Construction materials and gages shall be determined and selected to meet the design life and operational capability requirements of MIL-N-83155 and of this specification. At no position in the system shall the exhaust gas be allowed to directly impinge on the concrete foundations.

3.10.1 Augmenter. The augmenter shall collect the exhaust gas stream and augmented cooling air associated with each engine specified in 3.4, as applicable, and shall provide an enclosed path for these gases from the secondary air and test stand enclosures to the exhaust muffler. The augmenter design shall take into account:

a. The range of engine exhaust nozzle centerline heights associated with each engine when mounted on the engine runup trailer

b. The different engine exhaust nozzle planes relative to the runup trailer

c. The runup trailer restraint system.

3.10.2 Exhaust muffler. The exhaust muffler shall provide an acoustically treated path for the exhaust gases passed from the augmenter. Blast plates, deflectors, or any other components of the muffler mounted in the gas stream shall be designed to permit replacement without dismantling the entire structure. The final direction of the exhaust gases from the muffler stack shall be vertical.

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Drainage positions shall be located to prevent the collection of water in and around the muffler and shall utilize the drainage outlets shown on Drawing 69F179. Personnel access panel(s) shall be provided in the exhaust muffler for interior inspection and maintenance of the exhaust muffler. The access panels shall be accessible from ground level and shall not affect the noise attenuation of the exhaust muffler.

3.11 Water cooling system. The water cooling system shall circulate and control the required cooling water for the noise suppressor systems. The water cooling system shall be remotely controlled from the control cab. The water flow rates selected shall be calculated to prevent unevaporated water from being blown out of the exhaust stack at the stabilized engine running conditions specified in 3.11.1. A warning system shall be provided to alert personnel if overheating of the noise suppressor system occurs.

3.11.1 Water flow rates - A/F32T-2 noise suppressor. The water flow rates for the A/F32T-2 noise suppressor shall be controlled by one of the following:

a. A control system utilizing a maximum three-step water flow system. The low water flow rate shall not exceed 75 gpm for engine military power levels and operation of the TF30 engine at zone 1 afterburner. The middle water flow rate shall allow the correct amount of water into the exhaust gas stream during operation of the TF30 engine at zones 2 and 3 afterburner. The high water flow rate, not to exceed 800 gpm, shall allow the correct amount of water into the exhaust gas stream during operation of the TF30 engine at zones 4 and 5 afterburner operation and the remaining engines specified herein during afterburner operation. All water flow rate steps shall incorporate a means of manually adjusting the water flow rates to match the different operating conditions associated with each engine to be used with the noise suppressor. The water flow rates shall not be manually adjusted while operating the same engine model dash number, but may be adjusted for operation with a different engine model dash number or engine model such as J57-43 to J57-21 engine or TF30-3 to J75-17 engine, respectively.

b. An automatic, modulating control system using water flow rates up to, but not exceeding, 800 gpm during operation of the engines specified in 3.4a at all engine power levels including afterburner power. Override controls shall be included in the control panel specified in 3.8.17 to provide manual control in the event of malfunction of the automatic system. The override controls shall provide water flow at the maximum flow rate.

3.11.2 Water flow rates - A/F32T-3 noise suppressor. The water flow rates for the A/F32T-3 noise suppressor shall be controlled by one of the following:

a. A control system utilizing one or two water flow rates not exceeding a total flow rate of 800 gpm, including supplementary flow, to permit alternate operation of the J57 and J79 engines specified in 3.4b at afterburner power

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levels. A supplementary water flow rate of 75 gpm may also be used during operation of the engines up to and including military power level. All flow systems shall incorporate a means of manually adjusting the water flow rates to match the actual operating conditions associated with each engine to be used with the noise suppressor. The water flow rate shall not be manually adjusted while operating the same engine model dash number, but may be adjusted for operation with a different engine model dash number or engine model such as J57-43 to J57-21 engine or J57-21 to J79-15 engine, respectively.

b. An automatic, modulating control system using water flow rates up to, but not exceeding, 800 gpm during operation of the engines specified in 3.4b at all engine power levels including afterburner power. Override controls shall be included in the control panel specified in 3.8.17 to provide manual control in the event of malfunction of the automatic system. The override controls shall provide water flow at the maximum flow rate.

3.11.3 Water cooling system components. The water cooling system shall include the following:

a. All component parts of the system mounted on the noise suppressor

b. All connections, cable, or wiring for connection to the noise suppressor control panel in the A/M37T-6 control cab (see 3.8.17) utilizing the external connector in the terminal box (see 3.8.17.1)

c. Visual and audio warning indication in the control panel to warn operating personnel if the water level in the tank reaches a predetermined level. The system shall be activated from the low water level alarm junction box specified in the pumphouse shown on Drawing 69F179.

d. All utility lines required for connections between the pumphouse, control cab, and noise suppressor.

e. All equipment for installation in the pumphouse. The supplied and installed system shall include either of the following (see 6.2):

(1) Utilizing an above-ground water storage tank:

(a) A flanged 6-inch-diameter pipe to match the Government-furnished water supply connection outside the pumphouse specified on Drawing 69F179

(b) A gate valve inside the pumphouse to isolate the water supply from the downstream equipment

(c) A line strainer located upstream of the pump. The strainer screen shall be manufactured from 40-mesh monel wire, or monel wire cloth, or approved

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equal, having a wire diameter between 0.010 and 0.0135 inch. A retaining frame assembly shall be supplied to effectively support the screen and to withstand a 20-psi differential pressure without rupturing. The pressure loss across the strainer at system design flow rates shall not exceed 2 feet of water. The screen shall be positioned to allow easy access and removal for inspection and cleaning.

(d) A water pump selected to meet the system requirements and to receive the water through a 6-inch diameter pipe. At 800 gpm, the pressure head shall be considered to be a minimum of 2 feet and a maximum of 12 feet.

(e) A 277/480-V, 3-phase, 60-Hz electric motor to drive the water pump, provided with a starting system that reduces surge effects in the power supply.

(f) The necessary pipelines and valves to direct and control the water flow. A bypass line, incorporating a pressure-operated check valve, shall be installed to permit the pump to idle without an unacceptable pressure buildup to the supply circuit.

(g) An air compressor (if required) to satisfy all compressed air requirements for any pneumatically operated water system control valves, warning systems, et cetera. The compressor shall be driven by an electric motor selected to operate on the 277/480-V, 3-phase, 60-Hz or 120/240-V, single-phase, 60-Hz electrical power supply. The air produced by the compressor shall be free of oil droplets. The air compressor pressure tank shall meet the requirements specified in ASME Unfired Pressure Vessel Code. Belt guards shall be provided for personnel protection.

(2) Utilizing an underground water storage tank located beneath the pumphouse:

(a) A short-coupled vertical turbine pump, selected to meet the system water flow rate requirements. This pump shall be mounted on the flanged opening of the underground water storage tank or sump within the pumphouse. The maximum lift of the pump shall meet the water supply requirements as shown on Drawing 69F179. The minimum sump water level shall be 4 feet 6 inches above the bottom of the sump.

(b) An electric motor to drive the vertical turbine pump and selected to operate on the 277/480-V, 3-phase, 60-Hz electrical power supply. A starting system that reduces surge effects in the power supply shall be incorporated.

(c) An air compressor (if required) to satisfy all compressed air requirements for any pneumatically operated water system control valves, warning systems, et cetera. The compressor shall be driven by an electric motor selected to operate on the 277/480-V, 3-phase, 60-Hz or 120/240-V, single-phase, 60-Hz electrical power supply. The air produced by the compressor shall be free of oil droplets. The air compressor pressure tank shall meet the requirements specified in ASME Unfired Pressure Vessel Code. Belt guards shall be provided for personnel protection.

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(d) All pipelines and valves to direct and control the water flow. A bypass line, incorporating a pressure-operated check valve, shall be installed between the pump output line and the storage tank or sump, to allow the pump to idle without an unacceptable pressure buildup in the supply circuit.

3.11.4 Protection from freezing. All water pipes and components located outside of the pumphouse shall be protected against freezing of residual water that may remain in the system after shutdown. The required protection at each installation shall be based on the minimum temperature expected at that location (see 6.2). A means of draining the entire system in the event of an extended period of shutdown shall also be provided.

3.11.5 Temperature and water flow rate measurement and recording. To provide a permanent record of the operating temperatures experienced at the critical location(s) in the exhaust muffler, and the water flow rates passing through the system, a temperature and water flow rate recording system shall be furnished. The system shall include:

a. A two-pen recorder, with a 12-inch circular chart, geared to provide one revolution in 2 hours. The recording chart shall be calibrated to indicate temperature measurement from 0° to $1,000^{o}F$ for one pen and water flow rates from 0 to 800 gpm for the second pen. An initial supply of 1,000 charts shall be provided with each noise suppressor system. The recorder shall be located in the control cab adjacent to the air conditioning inlet duct and shall be braced from the air conditioning duct as the interior wall of the control cab cannot support more than 12 pounds. A separate or adjustable bracket shall be provided to mate with the A/M37T-6 and -6A and A/M37T-6B and -6C engine test stand control cabs. The recorder system accuracy, including transmitter, shall be $\pm 15^{o}F$ and ± 30 gpm from 0 to 400 gpm water flow and ± 15 gpm from 400 to 800 gpm water flow.

b. Measurement, by thermocouple or resistance bulb, of the temperature at the critical location(s) in the exhaust muffler. The electrical signal generated by these devices shall be used to operate one pen of the recorder.

c. A water flow rate measuring system to indicate the flow through the exhaust cooling system. The signal generated by the measuring device shall be used, or modified for use, with the second pen of the recorder.

d. A warning system shall be provided to alert operating personnel when an overtemperature condition exists inside the exhaust muffler. The system shall also alert personnel when there is a failure of the water cooling system or the electrical power supply. The warning system shall consist of a horn and flashing warning light mounted externally above the observation windows on

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top of the test stand enclosure sidewalls. A parallel circuit incorporating an audio signal (buzzer) and a warning light shall be installed in the control cab control panel. The control panel shall incorporate a switch(es) to test the operation of the overtemperature, water flow rate, and electrical power loss warning systems.

3.12 Pumphouse. A pumphouse of sheet metal modular construction and minimum size 12 by 20 by 8 feet high shall be provided and installed as shown on Drawing 69F179. All components located within the pumphouse shall be easily accessible for maintenance purposes from the access door.

3.12.1 General design. The pumphouse design shall incorporate the following essential features:

a. The building shall permit rigid attachment to the flat concrete base and shall withstand a wind and roof loading of 30 psf

b. Access shall be through double doors, providing an unobstructed opening 5 feet 8 inches wide and 6 feet 8 inches high. The door fastener shall permit the use of a padlock to secure the building.

c. Each wall shall include one window. Adequate ventilation shall be provided in the walls or roof. Where one window will be blocked by equipment, the window shall be moved to one of the other walls.

d. Insulation shall be installed, conforming to an overall U-value of 0.27 for the walls and 0.22 for the roof. Thermostatically controlled electrical heaters shall be provided to maintain an interior temperature of 40°F, based on a minimum exterior ambient temperature of -40°F.

e. Gutters and downspouts shall be provided on the pumphouse.

f. An access hole with an easily detachable cover shall be provided in the roof for installation or removal of the vertical turbine water pump with a boom crane. An access hole shall not be provided when a centrifugal water pump is provided.

3.12.2 Equipment and services. The following equipment and services shall be provided in the pumphouse:

a. Lighting fixtures, switches, and circuits to permit adequate illumination of all installed items for maintenance and inspection purposes. Three duplex female receptacles shall be provided on the inside of each sidewall and one duplex female receptacle shall be provided on the inside of each end wall for 120-V, 60-Hz, single-phase, 20-amp service. The receptacles shall conform to W-C-596, style X.

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b. All distribution boxes, disconnect switches, and wiring required to accept and distribute the electrical power supply to the various system components. The 277/480-V, 3-phase and the 120/240-V, single-phase electrical power supplies incorporate fused disconnect switches located outside the pumphouse, and shall be brought into the distribution box(es) in the pumphouse

c. A Type MC-1A motor generator to provide the 120/208-V, 400-Hz, 3-phase and 28-V, d-c power for the auxiliary equipment and control cab trailers

d. The water pumping, straining and control system equipments, et cetera, including all piping and wiring as necessary to satisfy the requirements of 3.11.3.

3.13 Supplementary fuel system. A 5,000-gallon fuel system for JP-4 turbine fuel conforming to MIL-T-5624 shall be provided to supplement the portable fuel system of the A/M37T-6 test stand. Fuel shall flow by gravity from the supplemental tank to the fuel tank located on the systems trailer of the A/M37T-6 test stand. The top of the supplemental fuel tank shall not be higher than the fuel tank mounted on the A/M37T-6 engine test stand systems trailer. Also, the bottom of the supplemental fuel tank shall not be lower than the bottom of the fuel tank mounted on the A/M37T-6 engine test stand systems trailer. The fuel outlet of the supplemental fuel tank shall be approximately 22 inches above ground level (see figure 4).

3.13.1 Supplemental fuel tank. The fuel tank shall be of the single-compartment type, having a minimum capacity for 5,000 gallons of JP-4 turbine fuel plus at least a 3 percent expansion space. The tank shall be elliptical or modified rectangular in cross section. The tank shall be constructed of 5,000 series sheet aluminum of sufficient thickness to provide complete structural integrity. The minimum sheet thickness shall be 0.187 inch for at least the bottom third of the tank. The remaining portion of the tank shall have a minimum thickness of 0.160 inch. The tank interior shall have a bright finish. Water-stained aluminum shall not be used for the tank. The tank shall be thoroughly neutralized and shall be cleaned inside and out to remove weld flux, scale, and other foreign matter. The tank shall be sloped to permit complete drainage by gravity through the sump. Rectangular-type tanks shall be reinforced and supported as necessary so that the top, bottom, and sides will not sag or be deformed. The tank shall be mounted on reinforced concrete saddles.

3.13.1.1 Components and fittings. All components and fittings shall be of nonferrous metals or 300 series stainless steel. Aluminum castings may be impregnated in accordance with MIL-STD-276 to prevent minor weeping. Elastomeric materials used in the tank, loading system, and all components which come into contact with the fuel shall be compatible with JP-4 turbine fuel conforming to MIL-T-5624.

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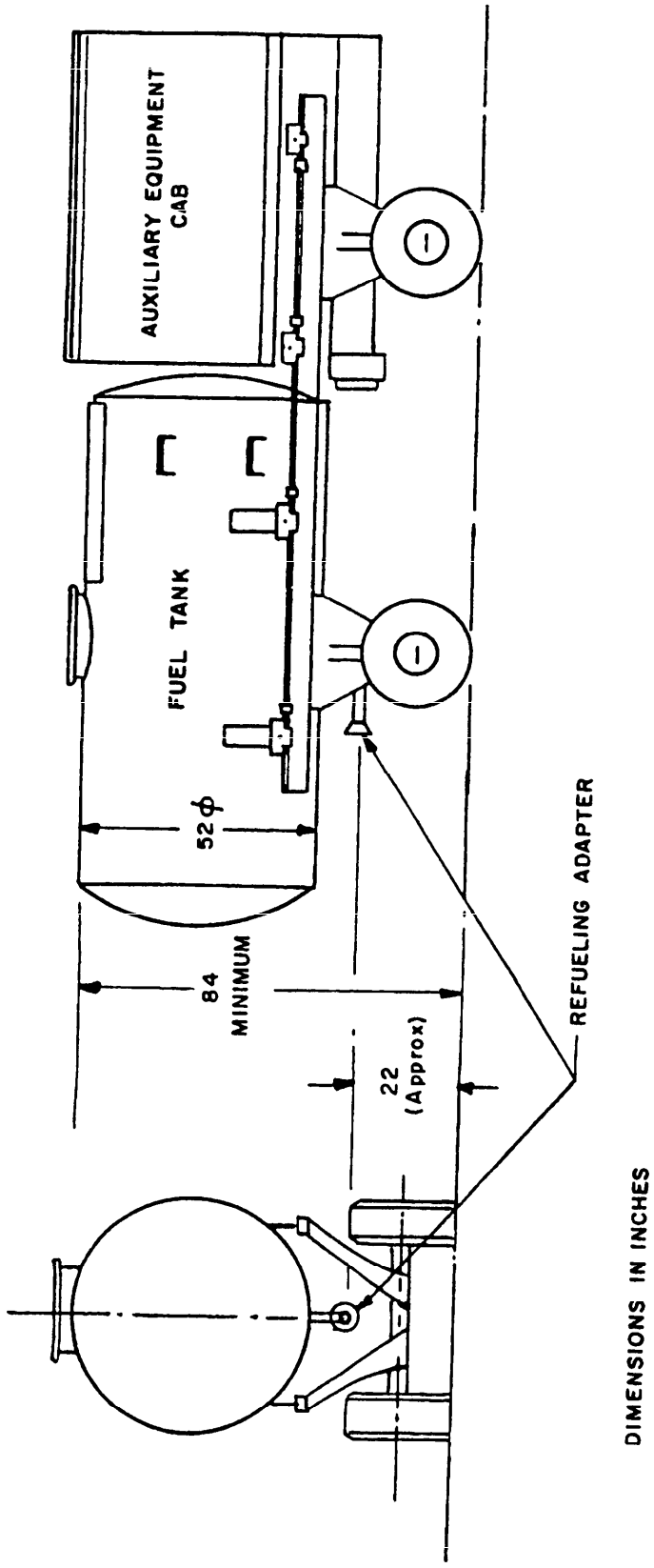


FIGURE 4. A/M37T-6 Fuel Tank Dimensions

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3.13.1.2 Sump. A sump shall be provided at the lowest point of the tank and shall have a minimum capacity of 12 gallons to allow accumulation of solid contaminants and water. The tank shall be so designed that water will drain into the sump and not stand in pockets.

3.13.1.3 Sump and tank drain. A 2-inch drain valve shall be installed in the sump to permit complete drainage of the tank and sump. The drain valve shall be in a protected location and shall be easily accessible.

3.13.1.4 Manhole. The tank shall have a circular manhole with a minimum diameter of 20 inches. The manhole shall have a 10-inch, watertight, hinged filler cover with necessary safety features to eliminate the possibility of a spark occurring during opening or closing of the cover. The manhole shall be equipped with a Buna N gasket, or equivalent, to insure a fuel-tight seal between the collar and cover. The entire top of the manhole shall be removable for entrance into the tank by removing a bolt-on clamping ring.

3.13.1.4.1 If baffling is provided in the supplementary fuel tank, a 20-inch-diameter manhole shall be provided in each baffle.

3.13.1.5 Vents. The tank shall be equipped with at least two identical removable vents. Each vent shall have sufficient capacity to relieve a 650-gpm liquid or vapor flow without allowing the internal tank pressure to exceed 2 psi. When fuel is flowing from the tank outlet, the vent shall remain open and not cause a vacuum of more than 1 psi below atmospheric pressure. The vents shall be equipped with an Underwriters' approved flame arrestor (see 6.3) and shall be designed to prevent ingress of rain.

3.13.1.6 Fuel level gage. A Boston Auto Gage Company, Incorporated model No. 114 tube-type gage, or equivalent, shall be provided to indicate the contents of the tank. The gage dial shall be white with black markings graduated in proportions of tank depth to indicate empty, one-fourth, one-half, three-fourths, and full levels in gallons. The gage shall be mounted in the bottom of the tank and so positioned that it can be easily read by personnel standing on the ground beside the bottom loading fuel inlet.

3.13.1.7 Steps and platform. The tank shall be equipped with a nonskid platform on top of the tank for servicing. Steps shall be provided on each side of the tank to permit access to the platform.

3.13.1.8 Bonding. Metal parts shall be bonded together to prevent the possibility of sparking.

3.13.2 Bottom loading system. The tank shall be equipped with a fully automatic bottom loading system. This system shall include an inlet manifold

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and a float-controlled, high liquid level shutoff valve. All components of the system shall be removable without entering the tank. There shall be no electrical equipment or electrical operation as a part of, or associated with, the bottom loading system.

3.13.2.1 Inlet manifold. The inlet manifold shall extend from the inlet receptacle to the bottom of the tank and shall terminate in the bottom of the tank. The manifold shall be a nominal 4-inch size and shall allow tank filling rates up to 650 gpm. The manifold and all its components shall be designed for static pressures up to 150 psi. The pressure drop through the manifold shall be kept to a minimum.

3.13.2.2 Inlet receptacle. The inlet receptacle shall be located at one end and below the bottom of the tank in a convenient location to allow for the attachment of the Type D-1 single-point nozzle and hose of the refueling vehicle. The receptacle shall incorporate a high-strength MS24484-2 adapter and a dust cover conforming to MIL-A-25896. The inlet receptacle shall be designed to open by attachment and operation of a Type D-1 nozzle. The receptacle and associated piping shall be substantially mounted to avoid failure due to the loads imposed by the nozzle, fuel-filled hose, or rapid increases and decreases in hose pressure.

3.13.2.3 High liquid level shutoff valve. A high liquid level shutoff valve shall be installed between the tank and the MS24484 inlet fitting. This valve shall be a diaphram-type valve with a single seat and composition disc. The valve shall utilize the line pressure for activation, and shall be so designed that failure of the diaphram will cause the valve to close. An arrow indicating the direction of flow shall be cast into the valve body and the valve shall have an external indicator to show the position of the valve seat. A self-cleaning, 40-mesh monel wire strainer shall be provided in the pressure supply line to the pilot valve. The valve opening and closing shall be controlled by a float-actuated pilot valve which is adjustable so that valve closure can be adjusted between 4,500 to 5,000 gallons of liquid in the tank. The closing time shall be between 3 to 6 seconds to prevent excessive surge pressures. The pilot valve with float and float rod shall be designed for installation and removal without entering the tank. The pilot control valve shall be a precision rotary disc, plate type valve. The diaphram valve and pilot valve shall have a design pressure rating of 150 psig. A metal box or other type structure shall be mounted in the tank, over the manifold outlet or the diaphram type valve, to prevent unrestricted spraying of the fuel when entering the tank.

3.13.3 Fuel transfer connection and hose. A fuel transfer connection, with a manually operated zero leakage shutoff valve shall be provided as the fuel outlet of the supplemental tank. A flexible hose, Aeroquip part No. 390A40D, or equal, shall be provided to connect the supplemental tank to the A/M37T-6 fuel tank. The hose shall be of sufficient length to connect the tanks when

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positioned as shown on Drawing 69E178. The hose shall be fitted with an adapter at one end, for connection to the fuel transfer connection, and shall have a single point Type D-1 nozzle on the other end for connection to the A/M37T-6 fuel trailer (see item A of figure 4).

3.14 Electrical system. The electrical system shall be designed to comply with Pamphlet 70.

3.14.1 Hazardous area. The areas enclosed by the primary air, test stand, and secondary air enclosures shall be considered as hazardous areas for electrical equipment that is operated during engine testing and all wiring or equipment provided and installed in these areas shall be in accordance with the provisions of Pamphlet 70, class 1, division 1, group D. The sealing of all penetrations into the enclosure areas through the equipment or foundations shall be completed in accordance with the requirements of Pamphlet 70, class 1, division 1.

3.14.2 Nonhazardous areas. All areas other than those defined in 3.14.1 shall be considered nonhazardous areas. The construction specifications contained in chapters 3 and 4 of Pamphlet 70 shall be incorporated into the system wiring and equipment provided and installed in the nonhazardous areas.

3.14.3 Electrical power source. Electrical power required to operate the noise suppressor system shall be taken from the commercial power source providing 150 kw of 277/480-V ±10 percent, 60-Hz ±5 percent, 3-phase and 20 kw of 120/240-V, 60-Hz, single phase at the junction box in the pumphouse specified in 3.12.

3.14.4 Grounding. All electrical equipment shall be grounded in accordance with Pamphlet 70. A plainly marked grounding connection shall be provided for connection to the grounding system.

3.14.5 Motors. All electric motors shall be of adequate rating to operate under all conditions at the maximum capacity required, without exceeding the manufacturer's recommended full load current.

3.14.6 Starters and starter cases. Electric motors rated at 5 hp or less shall be provided with across-the-line magnetic starters. Electric motors rated above 5 hp shall be provided with magnetically controlled, autotransformer-type, reduced-voltage starters. Transformer taps of 50, 65, and 80 percent of line voltage shall be provided. A single starter may be used if an automatic control system is provided. The control system shall give the operator the option of choosing the motor starting sequence. Interlocks shall be provided to prevent attempts to start more than one motor at a time, and to prevent switchover during acceleration of the first motor. The rating of the reduced

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voltage automatic control starter shall be selected based on the rating of the largest motor in the system. Starter cases of sufficient size to enclose all controls (contactor, overload devices, transformers, and starters) for all electric motors shall be furnished. The starters shall be capable of remote operation from the control panel in the control cab and shall include overload relays for motor protection that can be manually reset without opening the case.

3.14.7 Current protective devices. All electric motors shall be provided with one current protective device in each ungrounded conductor.

3.14.8 Wiring. All wiring between electrical components shall be in accordance with Pamphlet 70 and MIL-W-5088.

3.14.9 Cable conductor identification. Conductors shall be numbered in accordance with MIL-W-5088. In shielded or jacketed conductors, the individual conductors shall be color-coded and the outer jacket numbered.

3.14.10 Cable and conduit fittings. Cable, conduit, and fittings shall be installed in accordance with Pamphlet 70.

3.14.11 Switches. Standard AN or MS toggle switches conforming to MIL-S-3950 shall be provided on the control panel(s). The toggle switches shall be of such design that inadvertent contact with operators will not change their positions.

3.14.12 Interconnecting cable assemblies. Interconnecting cable assemblies required between the noise suppressor components shall be in accordance with MIL-C-3432, class HO or HD, type F flexibility.

3.14.12.1 Breakouts. At the breakouts, material conforming to MIL-S-8516, with a carbon-black filler added, shall be molded around the point where the conductors branch out. The molding shall provide support and seal the junction against moisture and dirt. The material shall have the same performance characteristics as the neoprene jacket. The configuration of the harness arrangement and type and position of the breakouts shall be neat, compact, and designed to perform the function for which it is intended. The cable breaking out shall have the same helically standard construction and the same flexibility as the cable before the breakout.

3.14.12.2 Connectors. Where not otherwise specified, connectors conforming to MIL-C-5015 shall be used throughout the noise suppressor system. All connectors exposed to the elements shall be provided with dust and moisture caps.

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3.14.12.2.1 Molded backshells. Backshells shall be molded with compound conforming to MIL-S-8516 with carbon-black filler added. The mold shall seal the back of the connector and bond to the jacket, eliminate the necessity for a clamp, and strengthen the junction between the cable and connector.

3.15 Fire protection systems. A primary and secondary fire protection system shall be provided in the test stand enclosure. The primary fire protection system shall consist of a Freon FE1301 or carbon dioxide fire fighting system. The secondary system shall be a water spray system.

3.15.1 Freon FE1301 fire protection system. If a Freon FE1301 fire protection system is selected, the amount of freon supplied shall be based on total flooding of the test stand enclosure from the roll-up door to the septum wall with a designed concentration of 6.6 percent plus a sufficient amount of Freon to maintain a concentration of 6.6 percent in the lower 2 feet from the engine inlet to the septum wall for 30 seconds under static conditions. The fire extinguishing nozzles shall be directed at the engine being tested. The Freon shall be stored in 100-pound containers on the test stand enclosure roof. This system shall be automatically activated by the temperature-sensing detectors specified in 3.15.3 and shall be designed in accordance with the standards of NFPA Bulletin 12A. The fire protection control panel shall be mounted on the control cab wall and shall have a maximum weight of 12 pounds. A means shall be provided on the control panel to manually activate this system.

3.15.1.1 Reserve supply of Freon FE1301. A reserve supply of Freon FE1301 shall be provided and connected into the distribution system specified in 3.15.1. The reserve supply shall have equal or greater capacity than the primary system specified in 3.15.1. The Freon shall be stored in 100-pound containers on the test stand enclosure roof. The reserve supply shall be manually controlled from either the fire protection control panel or the noise suppressor control panel.

3.15.2 Carbon dioxide fire protection system. If a carbon dioxide fire protection system is selected, the amount of carbon dioxide shall be 150 percent of the required amount for total flooding the test stand enclosure from the roll-up door to the septum wall under static conditions. The nozzles for the initial shot shall be directed at the engine being tested. Thirty percent of the initial carbon dioxide shot shall be arranged as an extended discharge system through separate piping and nozzles designed to discharge for 60 to 90 seconds in the 10-foot area forward of the septum wall. The system shall be automatically activated by the temperature-sensing detectors specified in 3.15.3 and shall be designed in accordance with the standards of NFPA-12. The fire protection control panel shall be mounted on the control cab wall and shall have a maximum weight of 12 pounds. Means shall be provided on the control panel to manually activate this system.

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3.15.2.1 Reserve supply of carbon dioxide. A reserve supply of carbon dioxide shall be provided and connected into the distribution system specified in 3.15.2. The reserve supply shall have an equal or greater capacity than the primary system specified in 3.15.2. The reserve supply shall be manually controlled from either the fire protection control panel or the noise suppressor control panel.

3.15.2.2 Carbon dioxide supply. The carbon dioxide supply shall be stored in 100-pound cylinders. The pumphouse shall be designed to provide space for storage of the cylinders. The area available for the other equipment shall not be affected. Sufficient access shall be provided to the cylinders for replacement, weighting, and maintenance. A weighing device shall be provided for weighing the cylinders.

3.15.3 Temperature-sensing detectors. Temperature-sensing detectors shall be provided for fire detection and activating the Freon or carbon dioxide fire protection system. The detectors shall be a combination rate-of-rise and fixed temperature and shall be spaced a maximum distance of 10 feet apart. The fixed temperature portion of the detector shall be activated at 180° ±15°F. Provisions shall be made for connecting into the base fire alarm system.

3.15.4 Water spray system. Separate water spray systems shall be provided in the test stand enclosure and secondary air intake enclosure. Each system shall be supplied water through deluge valves downstream of the noise suppressor water pump. Each system shall be designed to provide a minimum of 0.50 gpm per square foot floor area with each spray nozzle having a maximum spray pattern of 100° when both systems are operating at the same time. The water spray system shall be designed in accordance with the standards of NFPA-13 and -15.

3.15.4.1 Fixed temperature-sensing detectors. Fixed temperature-sensing detectors shall be provided for fire detection and activating the water spray system. The detectors shall be activated at a temperature of 280° ±15°F and shall be spaced a maximum distance of 10 feet apart. Separate detector systems shall be provided for the test stand enclosure and secondary air intake enclosure.

3.15.4.2 Water supply. The water supply for the water spray system shall be obtained from the noise suppressor water pumping system. An electric circuit shall be provided to activate the water pump in the event the water pumping system is not operating. Drainage provisions shall be incorporated to completely drain the water spray systems. Water shall not be supplied to the noise suppressor while the water spray is activated.

3.15.4.3 Provisions shall be incorporated to manually activate the water spray system from the noise suppressor control panel.

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3.16 Plumbing. All lines, valves, and fittings shall be of sufficient size to permit operation under maximum flow conditions. All lines, ducts, and tubing shall be identified in accordance with MIL-STD-1247.

3.16.1 Rigid lines. Commercial pipe and fittings may be used for the water system. Rigid lines for fuel, pneumatic, and oil systems shall be of corrosion-resistant steel or aluminum alloy. Rigid lines, fittings, and components shall withstand 200 percent of the maximum operating pressure without leakage or permanent distortion. Thread lubrication or teflon tape shall be used for sealing threaded joints.

3.16.2 Flexible lines and fittings. Flexible AN or MS hose with AN - or MS - type fittings shall be used. Hose clamps shall not be used for fluid line couplings. All fluid lines, valves, and fittings shall be of a type rated by their manufacturers for the pressure and service to be encountered and shall withstand 200 percent of maximum operation pressure without leakage or permanent distortion.

3.16.3 Interconnecting hose assemblies. The interconnecting hoses shall conform to the following:

a. Low and medium pressure (except waterline hoses) MS28741

b. High pressure MS28759

3.17 Cleaning, painting, plating, and anodic films. The cleaning, painting, and plating of all components of demountable noise suppressor systems shall be in accordance with MIL-STD-808. Components or parts of the noise suppressor systems that are made of stainless steel shall not require protective coatings. The finish colors of various components shall be as follows:

a. Primary air intake silencer
   Exterior - Gloss yellow, film designation DG
   Interior - Unpainted, galvanized, surface finish in accordance with finish No. P-116
b. Primary air intake enclosure
   Exterior - Gloss yellow, film designation DG
   Interior - Lusterless insignia white, film designation SL or galvanized
c. Secondary air intake enclosure
   Exterior - Gloss yellow, film designation DG
   Interior - Lusterless insignia white, film designation SL
d. Secondary air intake silencer
   Exterior - Gloss yellow, film designation DG
   Interior - Lusterless insignia white, film designation SL or unpainted, galvanized, surface finish in accordance with finish No. P-116

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20-amp service. The receptacles shall conform to [illegible]

e. Exhaust muffler and augmenter
   Exterior - Heat-resistant primer and finish with lusterless black silicone paint resistant to temperatures from 300° to 800°F (two coats)
   Interior - High temperature aluminum paint (two coats)
f. Pumphouse
   Exterior - Lusterless sand, film designation BL or standard color used by sheet metal building supplier if approved by the procuring activity
   Interior - Lusterless insignia white, film designation SL or standard color used by sheet metal building supplier if approved by the procuring activity
g. Engine test stand enclosure
   Exterior - Gloss yellow, film designation DG
   Interior - Lusterless insignia white, film designation SL (enclosure area forward of engine inlet may be galvanized)
h. Raceways
   Outside of enclosure - Gloss yellow, film designation DG
   Inside of enclosure - Lusterless, insignia white, film designation SL
i. External water pipes
   Lusterless black, film designation UL
j. Covered electrical outlets, switches, and circuit-breaker boxes
   Exterior - Gloss, insignia blue, film designation HG
k. Electrical power conduits
   Outside of enclosure - Gloss yellow, film designation DG
   Inside of enclosure - Lusterless, insignia white, film designation SL
l. Final painting shall be accomplished after the system is assembled and installed.

3.18 Markings

3.18.1 Stenciling. The following information shall be stenciled in 2-inch insignia red, film designation BG, letters on each side of the overhead hoist rail in a minimum of 2 places:

MAXIMUM HOISTING LOAD
2000 POUNDS

3.18.2 The following information shall be in red reflectorized tape, 6-inch letters on two opposite sides of the supplemental fuel tank:

FLAMMABLE
NO SMOKING WITHIN 50 FEET

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3.18.3 The following information shall be stenciled on or adjacent to the electrical power outlet switch specified in 3.8.9.1. The letters shall be 1 inch high, insignia red, film designation BG.

OFF DURING
ENGINE TEST

3.18.4 Electrical power outlets. The electrical power outlets specified in 3.8.9 shall be identified by means of a permanent plate.

3.18.5 Cable conductor identification. Conductors shall be numbered in accordance with MIL-W-5088. In the event of shielded or jacketed conductors, the individual conductors shall be color-coded and the outer jacket numbered.

3.18.6 Cable and hose identification. Cable and hose assemblies shall be identified with nondestructible, nonremovable, nonmetallic tags inscribed with the cable and hose number mounted in place at each end of the hose and cable not less than 3 inches from the connector.

3.19 Government-furnished property. When specified in the contract (see 6.2), the Government will furnish the following to the contractor upon his request:

Item No.	Item Description	Quantity
1	Load Bank, 60-kw, 40-KVAR, 3-phase, 400-cycle AC, Type A-1, MIL-L-4709	1
2	Compressor, Air, Gas-Turbine-Driven Trailer-Mounted, 150-Air-HP, Type MA-1, MIL-C-4851 or	1
	Generator Set, Power Unit A/M32A-60, MIL-G-38195	1
3	Motor-Generator, Mobile, Type MC-1A, MIL-M-9397	1
4	Test Stand, Aircraft Engine A/M37T-6, MIL-T-26289	1
5	Engine Control Kits and Adapters - to test engines specified in 4.5 (see 6.4)	As Req'd
6	Concrete foundations (see 6.2)	As Req'd.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the noise suppressor systems shall be in accordance with MIL-N-83155 and as specified herein.

4.3 Preproduction tests. The preproduction tests shall be performed on the first noise suppressor system delivered and installed and shall consist of all tests in MIL-N-83155 and all tests described herein (see 6.2).

4.4 Acceptance tests. The acceptance tests, to be performed on all noise suppressor systems delivered subsequent to the first system, shall consist of the individual tests in MIL-N-83155 and 4.5.4 through 4.5.13 herein.

4.5 Test methods

4.5.1 Acoustical evaluation. The acoustical performance of the noise suppressor systems shall be evaluated as specified in MIL-N-83155 using the following engines:

a. A/F32T-2 noise suppressor
   (1) TF30-P-7, P-12, or P-100 dash No. turbofan engine
   (2) J79-GE-15 or GE-17 dash No. turbojet engine

b. A/F32T-3 noise suppressor
   (1) J79-GE-15 or GE-17 dash No. turbojet engine.

4.5.2 Effect on engine performance tests. The effect on engine performance tests specified in MIL-N-83155 shall be performed with one of each of the following engines:

a. A/F32T-2 noise suppressor
   (1) J57-P-19, P-29, or P-43 turbojet engine
   (2) TF30-P-3 or P-9 turbofan engine
   (3) TF30-P-7, P-12, or P-100 turbofan engine
   (4) J79GE-15 or GE-17 turbojet engine

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(5) TF41-A-1 turbofan engine
(6) J75-P-5, P-17, or P-19 turbojet engine

b. A/F32T-3 noise suppressor
(1) J57-P-19, P-29, or P-43 turbojet engine
(2) J79-GE-15 or GE-17 turbojet engine

Where it is not practicable to run the engines on unsuppressed test stands as specified in MIL-N-83155, calibrated engines with documented performance data will be provided by the procuring activity for the effect on performance tests (see 6.2).

4.5.3 Durability test. The durability test specified in MIL-N-83155 shall be performed using:

a. A TF30 turbofan engine for the A/F32T-2 noise suppressor. The 4 hours minimum total operating time shall include 2 hours of afterburner power levels. The engine operations shall be arranged to include:

(1) Two separate continuous operations at military power for 45 minutes each and 10 operations at zone 5 afterburner power of 5 minutes duration, each separated by the minimum time permitted by applicable engine orders or the time required to replenish the noise suppressor water storage system, whichever is the greater.

b. A J79 turbojet engine for the A/F32T-3 noise suppressor. The 4 hours minimum total operating time shall include 2 hours at maximum afterburner power. The engine operations shall be arranged to include:

(1) Two separate continuous operations at military power of 45 minutes duration

(2) Ten operations at maximum afterburner power of 5 minutes duration

(3) Ten operations at maximum afterburner power of 1 minute duration and separated by a cooling period of 5 minutes duration.

4.5.4 Temperature and water flow rate measurement and recording test. The temperature and water flow rate measurement and recording systems, in compliance with the requirements of 3.11.5, shall be tested and check calibrated prior to use of the noise suppressor system with the test stand mounted engines. The check calibration results shall be included in the test report.

4.5.5 Hoist assembly test. The hoist assembly shall be tested to verify the hoisting requirements specified under 3.8.4.

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4.5.5.1 Hoist trolley and rail proofload test. The hoist trolley and rail shall be proof-tested by supporting a 4,000-pound load from the hoist trolley and transferring this load from one end of the rail to the other end of the rail. Any failure, permanent deflections, or yielding shall be cause for rejection of the noise suppressor system.

4.5.6 Roll-up door test. The roll-up door shall be tested to verify the operating requirements specified under 3.8.3.

4.5.7 Intercommunication extensions test. All extensions of the intercommunication system, installed inside and outside the test stand enclosure to meet the requirements of 3.8.8, shall be tested to insure correct operation.

4.5.8 Septum door adapter check. All adapters provided in satisfaction of the requirements of 3.8 shall be installed and checked to insure compatibility with the referenced test stand mounted engines.

4.5.9 Warning system tests. All warning systems specified in section 3 shall be tested to verify their required operation.

4.5.10 Supplementary fuel system tests

4.5.10.1 Inlet receptacle mounting. A Type D-1 single-point nozzle with a 100-pound weight attached by a cable shall be connected to the inlet receptacle. The weight shall be raised, and then dropped vertically for a distance of 4 $\pm$1/4 inch, and shall be stopped by the cable. This shall be repeated six times. There shall be no visible damage or indication of impending failure at the conclusion of the drops.

4.5.10.2 Inlet manifold hydrostatic test. The inlet manifold assembly, including the diaphram control valve, shall be subjected to a hydrostatic pressure of 150 psi for 10 minutes. There shall be no leakage in any portion of the manifold, including the closed valve, as a result of this test.

4.5.10.3 Static pressure test. The tank shall be filled with 5,000 gallons of JP-4 fuel and then pressurized with air to a static pressure of 3 psi for not less than 30 minutes. Evidence of leakage shall be cause for rejection.

4.5.10.4 Liquid level shutoff test. The tank shall be filled to capacity at a flow rate of 650 gpm through the inlet receptacle. The liquid level shutoff control shall close the manifold valve in not less than 3 seconds nor more than 6 seconds. This test shall be repeated five times, and the level of fuel in the tank shall not vary more than $\pm$1 inch from the preset level. At least 1,500 gallons of fuel shall be drained from the tank between each test run.

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4.5.10.5 Fuel transfer connection test. The tank shall be filled with 5,000 gallons of JP-4 fuel and allowed to stand for 30 minutes with the manual shutoff valve in the closed position. Evidence of leakage through the valve or connection shall be cause for rejection.

4.5.11 Fire protection system test. The correct triggering and operation of the fire protection systems specified in 3.15 shall be demonstrated. The extinguishing agent need not be expelled. Each temperature-sensing detector shall be tested for proper operation.

4.5.11.1 Extinguishing agent supply. Calculations shall be provided to verify that the amount of Freon FE1301 or carbon dioxide provided will completely flood the test stand enclosure when the engine being tested was operating at military power prior to release of the extinguishing agent. Calculations shall also be provided to verify that the water supply to the water spray system will provide a minimum of 0.50 gpm per square foot area.

4.5.12 Lighting tests. Lighting tests shall be conducted to verify the lighting requirements specified on 3.8.12 through 3.8.14.1.

4.5.13 Air velocity and temperature tests. Tests shall be conducted to verify that the air velocities and temperatures specified in table I are not exceeded. During these tests, it shall be demonstrated that the test stand enclosure does not have a depression greater than 2 inches of water as required by 3.4.1.

4.6 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Disassembly. Items shall be disassembled to the extent a substantial reduction in cubage is accomplished. All fasteners (bolts, nuts, pins, etc.) removed during disassembly shall be secured in one of the mating parts.

5.1.1.2 Matchmarking. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. Matchmarking information shall be on cloth shipping tags waterproofed in accordance with MIL-STD-129.

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5.1.1.3 The air intake enclosures, test stand enclosure, and pumphouse shall be preserved and packaged in accordance with MIL-P-116, method III.

5.1.1.4 The augmenter and exhaust muffler shall be individually preserved and packaged in accordance with MIL-P-116, method III.

5.1.1.5 The fuel tank shall be preserved and packaged in accordance with MIL-P-116, method I. The interior of the tank shall be fogged or sprayed with type P-10 preservative. All openings shall be plugged or capped. The tank shall be mounted on a skid-type base conforming to MIL-C-104. Contoured and fitted wood supports shall be provided. The tank shall be secured to the base with 1-1/4-inch flat steel strapping conforming to QQ-S-781. Further overboxing is not required.

5.1.1.6 The water pump shall be preserved and packaged in accordance with MIL-P-116, method I with P-2 preservative on all exposed bare metal.

5.1.1.7 The electrical system, floodlights, and miscellaneous equipment shall be preserved and packaged in accordance with MIL-STD-794. Selection of preservation and packaging methods shall be in accordance with appendix D, MIL-STD-794.

5.1.2 Level C. Each noise suppressor shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice if it meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. The air intake enclosures, test stand enclosure, and pumphouse shall be packed in containers conforming to MIL-C-3774, type 1, style A. Size and weight limitations of the crate shall not be exceeded. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.1.1 The augmenter and exhaust muffler shall be packed in containers conforming to MIL-C-3774, type 1, class A. Size and weight limitations of the crate shall not be exceeded. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.1.2 The water pump shall be packed in a container conforming to MIL-C-104, type 1, class 2, style A. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.1.3 The electrical system, floodlights, and miscellaneous equipment shall be packed in containers selected from tables I, II or III of MIL-STD-794. Size

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and weight limitations of the selected containers shall not be exceeded. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.2 Level C. The noise suppressor shall be packed in a manner that affords adequate protection against damage during shipment from the supply source to the first receiving activity. Containers shall conform to carrier rules and regulations as applicable to the mode of transportation.

5.3 Physical protection. Interior cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior containers shall be marked in accordance with MIL-STD-129. Set or assembly markings in accordance with the paragraph of MIL-STD-129 entitled Special Markings shall be applied to the containers.

6. NOTES

6.1 Intended use. The A/F32T-2 noise suppressor system is intended for use in reducing the noise generated during performance testing of the J57, J75, J79, TF-30, and TF-41 turbojet and turbofan engines when mounted on an A/M37T-6 engine test stand. The A/F32T-3 noise suppressor system is intended for use in reducing the noise generated during performance testing of the J57 and J79 turbojet engines when mounted on an A/M37T-6 engine test stand.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type noise suppressor system required (see 1.2)

c. Description of the Government-furnished facilities at the installation site (see 3.4)

d. Description of the Government-furnished electrical power supplies installation site

e. Details on installation of the A/M37T-6 engine run-up trailer and the modified restraint system (see 3.8.5)

f. Procedures for obtaining atmospheric temperature data for airbases (see 3.11.4)

g. Provisions for Government-furnished property (see 3.19)

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h. Location and conditions for preproduction testing (see 4.3), including:

(1) Extent of Government participation in testing

(2) Number and types of engines to be used for tests

(3) Certifications to be included in test report

i. Location and conditions for carrying out an unsuppressed acoustical survey on test stand mounted engines (see 3.6.1.1.2 and 4.5.2)

j. What septum door(s) will be furnished with the noise suppressor system (see 3.8.18.2).

k. What engine adapter tubes will be furnished with the noise suppressor system (see 3.8.18.3).

l. If the water storage tank is to be above ground or underground (see 3.11.3.e).

m. Selection of applicable level of preservation, packaging, and packing (see 5.1 and 5.2).

n. Special shipment marking (see 5.4).

6.3 Certification

6.3.1 Electrical certifications. The contractor shall submit proof that the electrical system and motors which will be supplied under 3.8, 3.11, 3.12, and 3.14 of this specification meet the requirements of Pamphlet 70 and the standards of Underwriters' Laboratories Incorporated with regard to fire and casualty hazards. The label or listing of the Underwriters' Laboratories Inc. will be accepted as evidence of conforming to the specification requirements.

6.3.1.1 In lieu of the label or listing, the contractor may submit independent proof satisfactory to the procuring activity that the equipment conforms to the requirements of Pamphlet 70 and the standards of Underwriters' Laboratories, Incorporated, with regard to fire and casualty hazards. Compliance with these requirements does not absolve the contractor from complete compliance with the other requirements of this specification.

6.3.2 Flame arrestor certification. The contractor shall submit proof that the flame arrestors specified in 3.13.1.5 meet the requirements of Pamphlet 70 and the standards of Underwriters' Laboratories Incorporated with regard to fire and casualty hazards.

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6.4 Reference material. The following documents are listed as sources of information in connection with this specification:

Technical Orders

1F-102A-10	Installation Testing Model J57-P-23A Engines
1F-105B-2-4	Power Plant, F-105B Aircraft
2J-J57-26	Turbojet Engines Model J57-P-23A, Performance Testing of
2J-J57-56	Turbojet Engines Model J57-P-43WB, Performance Testing of
2J-J79-46	Turbojet Engines Model J79-GE-15, Performance Testing of
2J-J79-56	Turbojet Engines Model J79-GE-17, Performance Testing of
2J-TF30-16	Aircraft Engines USAF Model TF-30-P-3, Performance Testing of
2J-TF30-26	Aircraft Engines USAF Model TF30-P-12, Performance Testing of
2J-TF41-6	Field Maintenance Instructions, USAF Model TF41 Engines
33D4-6-101-1	J79-GE-15 Bare and QEC Engine Control Kit
33D4-6-115-1	J79 Bare and F104A, B, C, D, and G, QEC Engine Control Kits
33D4-6-212-1, -11	Operation and Maintenance for Aircraft Jet Engine Test Stand, Type A/M37T-6
33D4-6-212-4	Illustrated Parts Breakdown for Aircraft Jet Engine Test Stand, Type A/M37T-6
33D4-6-88-1	Operation Service A/M37T-1, 1A
33D4-6-249-1	TF30-P-1 Bare and QEC Engine Control Kit

6.5 Definitions

6.5.1 A/M37T-6 engine test stands. The A/M37T-6 engine test stand shall include all A/M37T-6 series engine test stands. These stands are the A/M37T-6, -6A, -6B, and -6C engine test stands.

6.6 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4920-F270

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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